

ADEM HMERCHA

Junior Software Engineer | Full Stack Developer

ademhmerchaaa@gmail.com • +216 50 733 444 • Tunis, Tunisia • github.com/ademhmercha

PROFESSIONAL SUMMARY

Junior Software Engineer and Full Stack Developer with hands-on industry experience designing and shipping production-grade web applications, real-time communication systems, and cloud-native infrastructure. Holds a Software Engineering degree from ESPRIT, Tunis, with two completed internships building enterprise platforms using the MERN stack, Docker, and Kubernetes. Experienced across the complete development lifecycle — from pixel-precise React interfaces and RESTful Node.js backends to automated CI/CD pipelines and Kubernetes auto-scaling. Proven track record with real-time technologies (Socket.io, WebRTC), AI-powered tooling, and full DevOps automation. Committed to writing clean, maintainable code and delivering solutions that solve real-world problems at scale.

EXPERIENCE

Software & DevOps Engineering Intern (PFE)

Jan 2026 – Jun 2026

Sagemcom • Ezzahra, Tunisia

- Architected and delivered **Telnet Test Manager**, an enterprise-grade platform for automating the testing and monitoring of network equipment, adopted by multiple cross-functional engineering teams across the organization.
- Implemented **bidirectional WebSocket communication** using Socket.io to stream live test execution updates and real-time status changes to connected clients, eliminating manual polling and significantly improving operational efficiency for test engineers.
- Designed a comprehensive **Role-Based Access Control (RBAC)** system with granular, department-level permissions, enabling secure and structured multi-team collaboration across shared test infrastructure without privilege conflicts.
- Built and maintained an end-to-end **CI/CD pipeline** using GitHub Actions, Docker, and Kubernetes — automating test execution, container builds, image pushes, and zero-downtime rolling deployments to production on every merge.
- Integrated **Prometheus and Grafana** to deliver full-stack observability: custom dashboards for API latency, pipeline health, and resource utilization, alongside alerting rules that proactively surface degraded services before user impact.
- Deployed and managed a production-ready Node.js REST API on a Kubernetes cluster, configuring **Horizontal Pod Autoscaler (HPA)** to dynamically scale from 2 to 10 pods based on live CPU utilization, ensuring responsive performance during peak load without over-provisioning.
- Optimized containerization using **multi-stage Dockerfiles** with non-root execution and graceful shutdown handling — reducing final image size, hardening the security posture, and ensuring clean process termination under production load.
- Authored PowerShell **load-testing scripts** with automated setup and teardown procedures to simulate production traffic, validate HPA scaling thresholds, and document system performance benchmarks for stakeholder review.

Software Development Engineering Intern

Jun 2025 – Jul 2025

Standard Sharing Software • Tunisia

- Contributed to the design and development of client-facing web features using React and Node.js, delivering responsive, accessible UIs aligned with product specifications and business requirements.
- Participated in agile sprint cycles — attending daily standups, code reviews, and retrospectives — gaining practical experience with collaborative software delivery in a professional engineering environment.
- Integrated third-party APIs and services to extend platform capabilities, documenting endpoints and behaviors in Postman to support ongoing team development and onboarding.

PROJECTS

Safeguard — Full-Stack Parental Monitoring Platform

React · Node.js · PostgreSQL · Supabase · Firebase FCM · React Native · Expo

- Built a cross-platform parental control system with a React web dashboard and React Native mobile app, featuring remote device lock/unlock, granular website blocking, screen time scheduling, and real-time push alerts via Firebase Cloud Messaging.

AutoReview Agent — AI-Powered Multi-Agent Code Auditor

React · Vite · Groq API · Llama 3.3 70B

- Developed a fully browser-based code auditing tool powered by 5 specialized AI agents (quality, security, architecture, documentation, performance) using Groq/Llama 3.3 70B — no backend required. Includes PDF report export for audit results.

Matchy — Real-Time Dating Application

React · Node.js · MongoDB · Socket.io · WebRTC

- Engineered a real-time dating platform with swipe-based matching, live Socket.io chat, WebRTC peer-to-peer video and voice messaging, and a multilingual EN/FR interface. Includes an admin dashboard with analytics, audit logging, and user management.

DRAPE — Luxury Fashion E-Commerce Platform

React · Node.js · MongoDB · Tailwind CSS · Cloudinary · Whapi.Cloud

- Built a full-featured e-commerce platform with JWT authentication, Cloudinary-powered image management across 4 product categories, a complete cart and checkout system, and automated WhatsApp order notifications via the Whapi.Cloud API.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, Tailwind CSS, React Native, Expo, Vite, i18next

Backend: Node.js, Express.js, MongoDB, PostgreSQL, Prisma, Supabase, Socket.io, WebRTC, JWT Authentication, REST APIs

DevOps & Cloud: Docker, Kubernetes, HPA, GitHub Actions, CI/CD Pipelines, Prometheus, Grafana, Vercel, Railway

Tools & Services: Git, Firebase (FCM), Cloudinary, Whapi.Cloud, EAS, Figma, Postman

EDUCATION

Bachelor of Engineering — Software Engineering

2023 – 2026

ESPRIT · Tunis, Tunisia

Preparatory Classes — Mathematics & Physics

2021 – 2023

IPEiM · Monastir, Tunisia

LANGUAGES

Arabic: Native

French: Professional proficiency

English: Professional proficiency